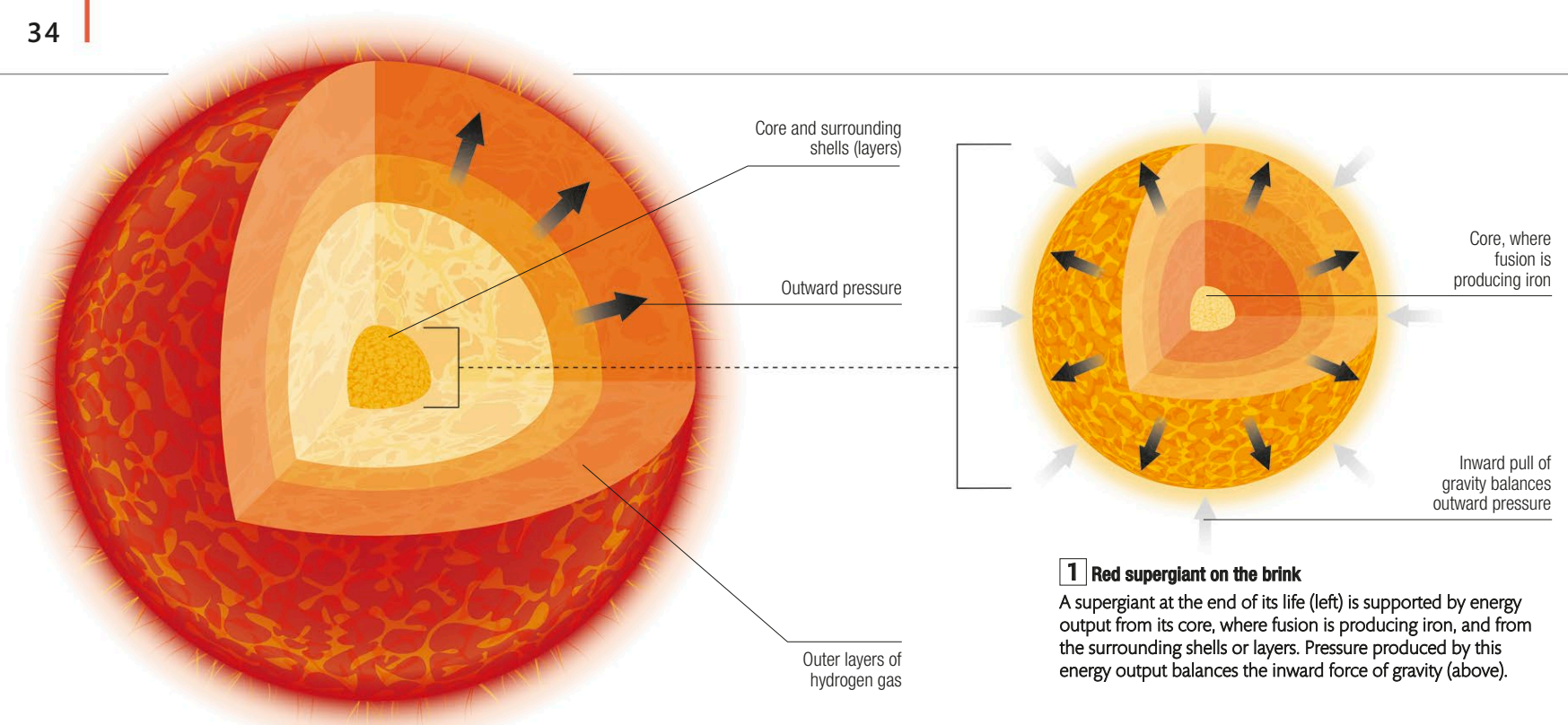




SUPERNOVAE

A SUPERNOVA IS THE CATAclysmic EXPLOSION OF, IN MOST CASES, A HIGH MASS STAR AT THE END OF ITS LIFE. A SUPERNOVA BLASTS OUT SO MUCH LIGHT AND OTHER ENERGY THAT IT CAN BRIEFLY OUTSHINE A GALAXY.

Supernovae are quite rare astronomical events in individual galaxies. None has been clearly observed in our galaxy since 1604, when a supernova some 20,000 light-years away was visible to the naked eye. However, a growing number of supernovae have been spotted in other galaxies, including one in the Large Magellanic Cloud (a satellite galaxy of the Milky Way) in 1987. A new, bright, supernova might occur in our galaxy at any time.

Types and causes

Supernovae are classified according to their spectra into various types, such as 1a, 1b, and II. Types II and Ib are the main varieties in which very high mass stars explode. As they reach the end of their life, these stars swell into supergiants and obtain their energy from nuclear fusion reactions going on in their cores and in a series of shells or layers surrounding their cores. Eventually they start making iron in their cores, but fuel for this process soon runs out. As iron itself cannot be fused to supply energy, energy output in the core suddenly ceases, and this triggers a massive explosion.

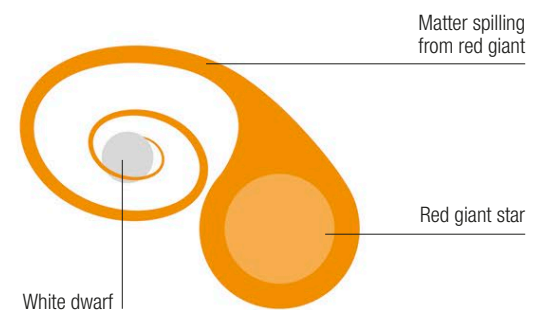
Some chemical elements can be forged only in the **extreme high-energy** conditions of a supernova

Type 1a supernovae

Although most supernovae are caused by the rapid collapse and violent explosions of very high mass stars, one type—known as a Type 1a supernova—has a different mechanism. Supernovae in this category occur in binary star systems (pairs of stars orbiting each other) where at least one star is a white dwarf (see p.20). The transfer of material from a companion star onto a white dwarf, or the collision of two white dwarfs, can both cause Type 1a supernova explosions. These explosions tend to have a uniform light output, which makes observations of them in distant galaxies useful for measuring the distances to those galaxies.

1 Matter transfer between orbiting stars

An aging star, which has swelled into a red giant, begins to spill some gas from its outer layers onto a white dwarf star it is orbiting. This can lead to bright outbursts, called novae, on the surface of the white dwarf.



2 White dwarf explodes

The white dwarf's mass gradually increases from the extra gas it is acquiring. Eventually it becomes unstable and explodes as a Type 1a supernova. The explosion may cause the red giant to be blasted away.

